Glass batch chemistry with high-purity aragonite

Representative chemistry and evaluation questions for modern glass formulations.

Summary

Calcium carbonate is a common CaO source in glass batch formulations. Purity, particle-size distribution, moisture and iron content can influence batch behavior and finished-glass requirements. High-purity natural oolitic aragonite is a candidate for documented trials in container, flat, solar, fiberglass and specialty glass formulations.

Why aragonite in glass batches

Batch chemistry Confirm $CaCO_3$, CaO contribution and impurity limits against the complete formulation.	Particle distribution Evaluate sizing, fines and moisture for mixing, dust handling and furnace conditions.
Iron and clarity Review current Fe_2O_3 data where glass color or optical quality is sensitive.	Lifecycle questions Do not assign an emissions benefit without a defined, independently reviewed lifecycle comparison.

Glass-grade specification snapshot

$CaCO_3$ ≥ 97.5%	Whiteness ≥ 92
Fe_2O_3 ≤ 0.04%	Typical mesh 100–325

Typical chemistry (raw material)

Constituent	Typical	Notes
$CaCO_3$	≥ 97.5%	Calcium carbonate, primary
$MgCO_3$	≤ 0.9%	Magnesium carbonate
SiO_2	≤ 0.6%	Silica
Fe_2O_3	≤ 0.04%	Iron oxide — low
Al_2O_3	≤ 0.10%	Alumina
$Na_2O + K_2O$	≤ 0.30%	Alkalis
LOI	~ 43%	Loss on ignition

Applications

Potential evaluations include container, flat, solar, fiberglass and specialty glass formulations. Batch performance depends on furnace design, cullet ratio, particle distribution and the full formulation. A technical specialist can help identify current documentation and a starting sample for customer-led trials.

Conclusion

For glass producers comparing $CaCO_3$ sources, aragonite's crystal form and oolitic morphology justify a documented batch trial. Final suitability should be based on current material data and results from the intended furnace and formulation.

This brief summarizes representative properties for comparative and educational purposes. Values are not guaranteed; request current specifications and available lot-specific documentation. info@aragocorminerals.com · +1 (209) 487-0110